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Resistance and embedment depth of embedded rectangular concrete-filled steel tubular column base connection under cyclic loads

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A R T I C L E I N F O

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A B S T R A C T

The column base connection, served as transferring the external loads acted on the upper structures to the foundation, its behavior is essential for the safety of constructions. The embedded column base connection is widely adopted in steel constructions, especially in high-rise buildings. Up to now, the studies on the seismic behavior of concrete-infilled steel tubular (CFST) column base connections have been very limited. In this study, two specimens with the relatively shallow embedment are tested under the lateral cyclic loading, and then a refined finite element (FE) model is developed. Using the verified FE model, a parametric study is conducted with the effects of main parameters taken into account. According to test and FE results, the embedded CFST column base connections may fail by the concrete bearing failure, the upward punching failure and the CFST column failure, depending largely on the embedment depth. A solution for predicting the ultimate load of the embedded CFST column base connections is proposed. Comparisons with FE results and predictions of existing method illustrate that the proposed solution achieves an average prediction accuracy of 0.95, which is superior to 0.39, 0.28 and 0.65 for the representative methods of GB 50017, JGJ 138 and Grilli. Subsequently, the required embedment depth is also presented by assuming the resistance of base connection not less than that of CFST column section. The embedment depths using the presented solution are approximately 0.58–0.63 times of the requirement embedded depths codified in Chinese design codes (e.g. GB 50017, JGJ 138), which is believed to be meaningful for the application of this type of column base connections.

1. Introduction

The behavior of the column base connection, served as the structural component for transferring external loads acted on the upper structure to the foundation, is essential for the safety of constructions. Column base connections in steel constructions are generally classified into three main types: exposed column base connection [1,2], encased column base connection [3] and embedded column base connection [4,5]. Of these column base connections, the embedded column base connection has been frequently adopted in practice, especially in high-rise steel buildings.

Past post-earthquake surveys have consistently identified column base connection failure as a predominant failure mode in steel structures. For instance, the investigation following the 1995 Hyogo-ken Nanbu earthquake revealed that numerous steel buildings suffered damage directly attributable to column base failures [6]. Therefore, research into the load-transfer mechanisms and design methods of column base connections is critically important for enhancing the safety and performance of steel structures.

Up to now, extensive investigations have been conducted on the behavior of embedded column base connections [7–12]. Among these studies, Hsu and Lin [9] investigated the seismic performance of embedded rectangular CFST column base connections with anchor bolts with/without special designed vertical stiffeners. They found that using deep embedment and setting the vertical stiffeners may effectively improve the stiffness and energy dissipating capability of embedded column base connections. Grilli *et al.* [10] tested five embedded H-column base connections and proposed a strength model based on a side-bearing mechanism incorporating the contributions from the extended end plate. Moon *et al.* [11] tested the circular CFST embedded column base and found that the embedment depth had the most significant impact on the failure mechanisms. Xu *et al.* [12] conducted an experimental study on the embedded rectangular CFST column base connections with enlarged endplate and stiffeners. Besides the studies on

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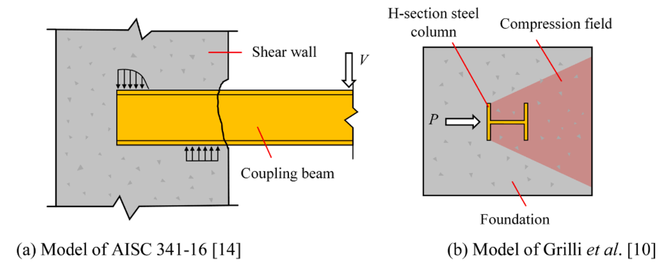


Fig. 1. Existing models for ultimate load prediction of embedded steel column base.

the behavior, the solutions for predicting the ultimate load of embedded steel column base connections have been also developed in existing studies (e.g. [10] and [12]) and design codes (e.g. [13] and [14]).

As explored by existing studies, a critical factor influencing the load-carrying capacity of embedded column base connections is the embedment depth. The minimal embedment depths are required in several design codes for steel or concrete-filled steel tubular (CFST) column connections embedded into concrete foundations. For instance, Chinese codes JGJ 138–2016 [15] suggests that the ratio of the embedment depth to the height of column section should not be less than 2.0, while, in JGJ 99–2015 [16], this ratio becomes 2.0 for steel column of H-section or 2.5 for box-section. In GB 50017–2017 [13], a formulation is given for the minimal embedment depth of embedded column base connections, based on the side-bearing concrete failure assumption.

In these days, CFST columns are more and more popular for use in high-rise steel buildings due to their superior structural performances over steel and concrete columns. However, the investigations on the seismic behavior of the embedded CFST column base connections have been very limited, up to now. Moreover, the existing solutions for ultimate load prediction of embedded steel column base connections may not be applicable to the CFST column base connections. This is because, for example, the AISC 341–16 [14] solution is an extrapolation of the coupling beam formulations (see Fig. 1a); the *V-M* interaction relationship in GB 50017–2017 [13] is based on the theoretical model for the side-bearing failure; and the solution of Grilli *et al.* [10] was developed according to the test results of H-section column base, which may be not applicable for rectangular-profiled sectional columns (see Fig. 1b). The practical applications have indicated that the minimal embedment depth required in design codes (e.g. [15] and [16]) may bring great difficulties for in-site installation of embedded steel/CFST column base connections, especially for columns with large sectional dimensions.

Based on the above discussions, CFST columns have been widely used in steel structures; however, the behavior of embedded CFST column base connections has not been systematically investigated. Most existing design approaches in the literature and current design codes are based either on side-bearing mechanisms or on experimental results of H-shaped steel columns, which may not be directly applicable to embedded CFST column bases. Therefore, this study aims to reveal the seismic performance of embedded CFST column base connections, and

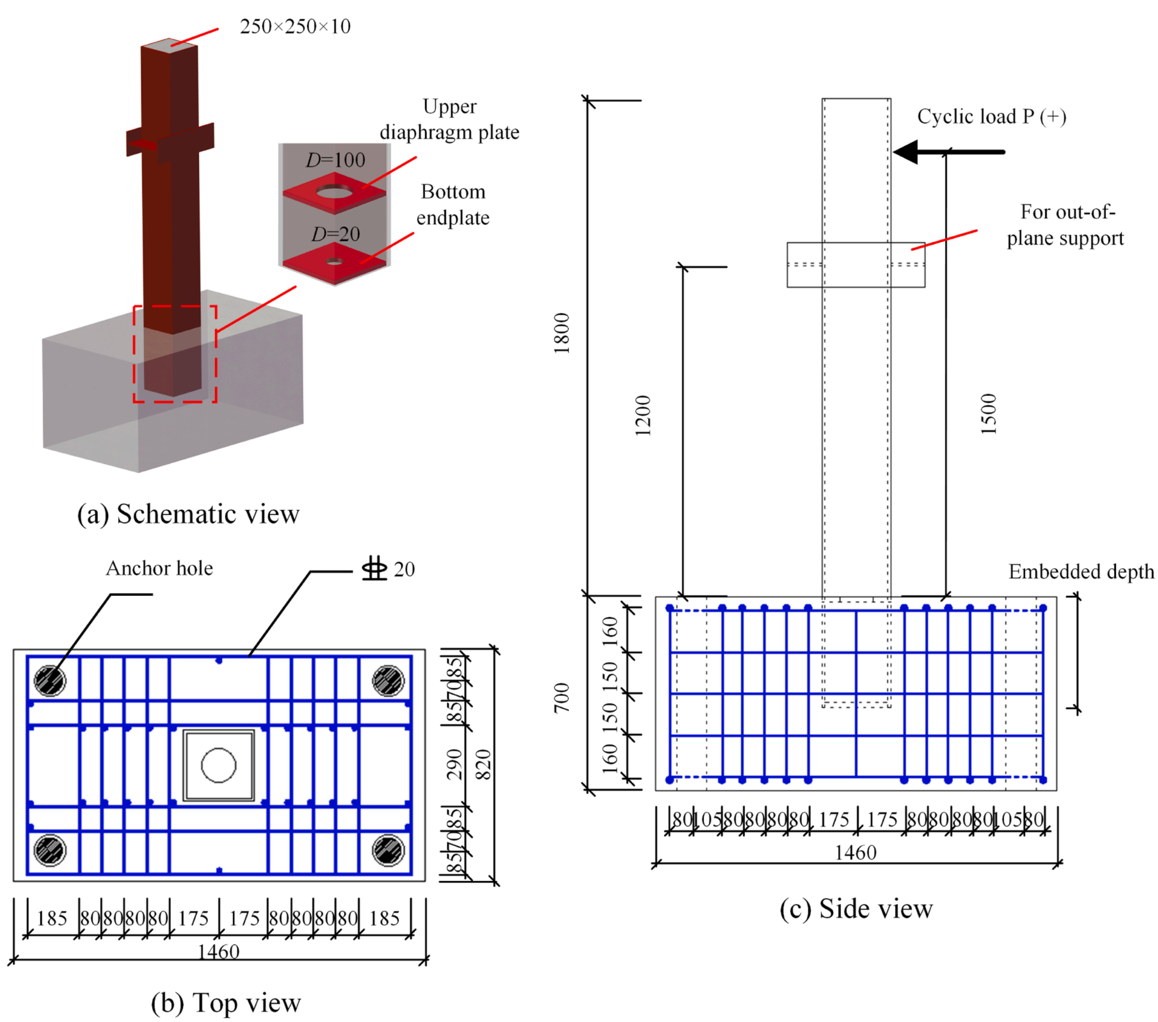


Fig. 2. Details of the test specimens (mm).

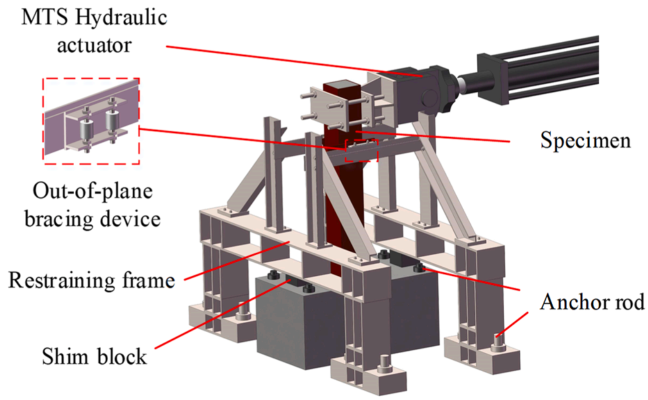


Fig. 3. Test setup.

subsequently present the specific methods for calculating the resistance and required embedment depth of embedded CFST column base connections. To this end, two specimens with the relatively shallow embedment are tested under the lateral cyclic loading. Based on the test results, a refined finite element (FE) model is developed and verified. The parametric study with effects of main parameters taken into account is then conducted employing FE analyses. Based on the test and parametric study, a solution for predicting the ultimate load of the embedded CFST column base connections is then proposed, with which the required embedment depth is also developed by using the similar assumption as GB 50017–2017 [13].

2. Experimental study

2.1. Test specimens and setup

Two embedded column base specimens with the embedment depths of 300 mm (ECB300) and 400 mm (ECB400) were tested under the quasi-static cyclic lateral loading. It should be noted that the embedment depths (300 mm and 400 mm) adopted for specimens are less than the typical embedment requirement (i.e. the embedment depth/the height of column section not less than 2.0) specified in design codes (e.g. JGJ 138–2016 [15]). This is because the failure within the embedded region of specimens was expected in this test, as the strength of CFST column has been extensively investigated in literature. On the other hand, according to the previous study [11], the embedment depth may not be less than 0.8 times the height of column section in practice.

As shown in Fig. 2, each specimen is composed of a rectangular concrete-filled steel tubular (CFST) column with the cross-section of 250 mm× 250 mm× 10 mm (depth×width×thickness), and a reinforced concrete foundation with the length, width and height being 1460 mm, 820 mm and 700 mm, respectively. Two 10 mm thick plates are set in the steel tube: the upper diaphragm plate is arranged at the height of top surface of concrete foundation featuring a 100 mm diameter circular opening for concrete pouring; and the bottom end plate is welded to the end of steel tube, with a 20 mm diameter vent hole. Different from the extended end plate configuration, the edges of the bottom end plate considered in this study are not beyond the contour of CFST section. The details of steel reinforcements in the concrete foundation are also shown in Fig. 2, where the ribbed steel rebars with diameter of 20 mm and yield strength of 400 MPa are used for vertical and horizontal reinforcements.

As shown in Fig. 3, the cyclic lateral loading was applied at the section with L_c= 1500 mm above the top surface of concrete foundation through the MTS hydraulic actuator, and the out-of-plane deflection is prevented via a bracing device setting at the height of 1200 mm. The loading height (1500 mm) used in the test is about the half-height of typical story-height of construction (i.e. 3000 mm), and the lateral load is assumed to be acted at the inflection point of column under lateral loads.

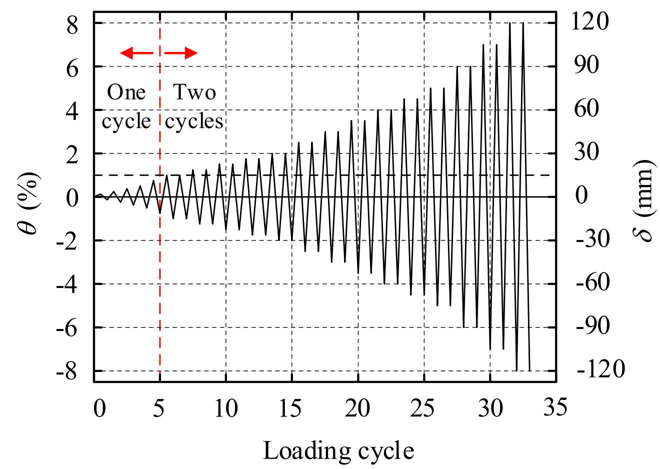


Fig. 4. Cyclic loading protocol.

The cyclic loading was performed according to the loading strategy illustrated in Fig. 4, which was designed according to JGJ/T 101–2015 [17]. A lateral load equal to 1/10 of the designed ultimate load was applied to the CFST column via the MTS Hydraulic actuator prior to the formal loading procedure to eliminate the gaps between different components and to ensure the proper functioning of the loading and measuring systems. The drift ratio θ is defined as Eq. (1).

$$\theta = \frac{\delta}{L_c} \tag{1}$$

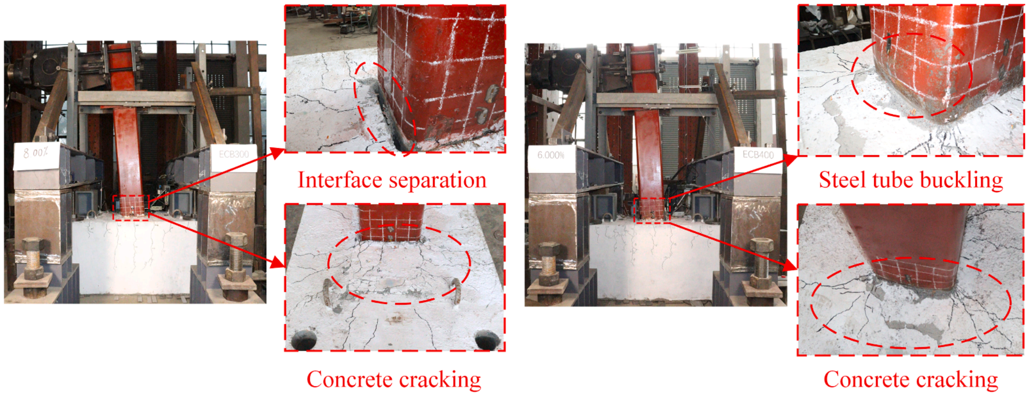
where δ and L_c are the lateral deformation and the distance between the concrete foundation and loading point, respectively.

2.2. Failure mode

Fig. 5a and b present the failure modes of the test specimens. For the ECB300 specimen, the severe separation between the CFST column and surrounding concrete was observed along the direction of the cyclic loading (Fig. 5a). As shown Fig. 5a, the concrete crushing happened near the interface between the CFST column and surrounding concrete, and a number of cracks also developed in the vicinity region along the loading direction. Different from ECB300, the specimen ECB400 failed by the local buckling of the outer steel tube CFST column, and no obvious interface separation was observed (Fig. 5b). The cracks were also found in ECB400 at the top surface of concrete foundation, mainly being the radiating diagonal cracks near column corners.

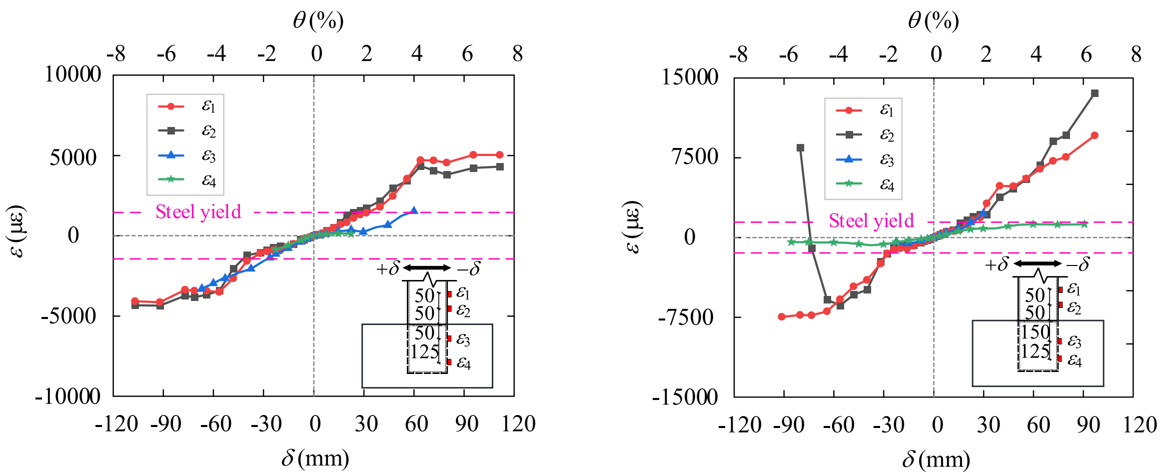
As shown in Fig. 6, four strain gauges (ε_1-ε_4) were attached to the column surface near the top face of concrete foundation, and the vertical positions of ε_1-ε_4 were + 100 mm, + 50 mm, −50 mm and −175 mm with respect to the top face of concrete foundation, respectively. It should be noted that the strain gauges were stuck to the outer surface of the mid-width of steel tube, which may lead to the measured strains being sensitive to the local bending deformation of steel tube.

It can be seen from Fig. 6 that, for both specimens, the longitudinal strains of two strain gauges below the concrete top surface (i.e. ε_3 and ε_4) are significantly less than those of ε_1 and ε_2, especially at δ> 15 mm. The outer steel tube began to yield at a drift ratio of about 1.6 % and 1.2 % for ECB300 and ECB400, where the yield strain of the outer steel tube, 0.00144, was determined using the coupon test results. The longitudinal strains of the two strain gauges above the top surface of concrete foundation (ε_1 and ε_2) gradually increased as the lateral displacement (δ). For the specimen ECB300 shown in Fig. 6a, the increases of ε_1 and ε_2 with δ were not obvious when δ was greater than about 60 mm, which is because the lateral load did not increase any more due to the severe



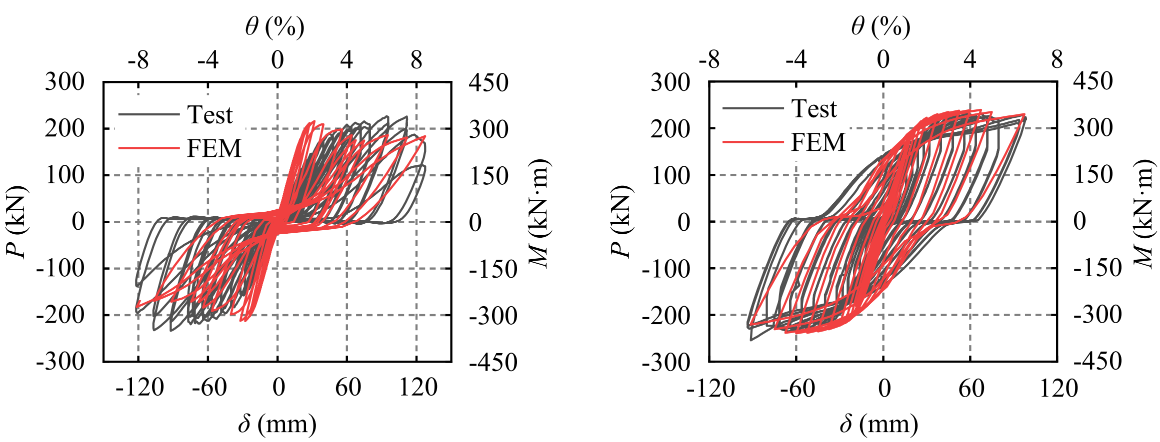
(a) ECB300 (b) ECB400

Fig. 5. Failure modes of specimens.



(a) ECB300 (b) ECB400

Fig. 6. Strain variations of specimens.



(a) ECB300 (b) ECB400

Fig. 7. Hysteresis curves of both specimens.

crushing of concrete happened. The similar tendency as ECB300 of ε_1 and ε_2 for ECB400 can be seen in Fig. 6b for δ< 60 mm, while at the loading stage of δ> 60 mm the strains of ε_1 and ε_2 were obviously affected from the local deformations of steel tube, leading to the sudden changes of ε_2 (see Fig. 6b).

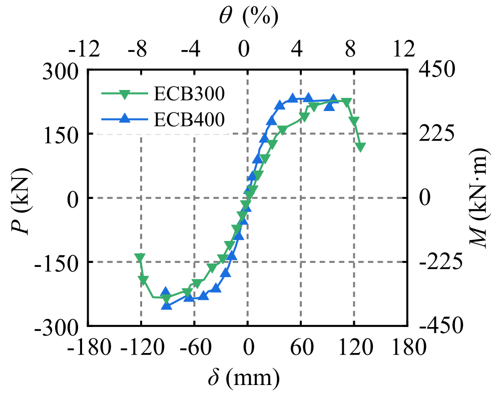


Fig. 8. Skeleton curves.

2.3. hysteresis behavior

Fig. 7 plots the hysteretic response characteristics of both specimens, where P and M are the lateral force and bending moment. Note that M and θ in Fig. 7 are both with respect to the column-foundation interface (calculated as $M = P \times h$ and $\theta = \delta/1500$, with h being the loading height 1500 mm). It can be seen from Fig. 7a that the severe pinching responses exhibit in the second and fourth quadrants of the hysteresis curve of ECB300, which is mainly due to the interfacial separation shown in Fig. 5a. In the contrast, the hysteresis curve of ECB400 is more fusiform, indicating its much better energy dissipation capacity. Moreover, the sliding phenomenon near the horizontal axis of the hysteresis curve of ECB400 is slight compared to ECB300, induced by the interfacial separation between CFST section and surrounding concrete (Fig. 5a).

2.4. Skeleton curve and energy dissipation capacity

The skeleton curves of both specimens are shown in Fig. 8, which are obtained by sequentially connecting the peak load points of each loop in the hysteresis curves. It can be seen from Fig. 8 that the skeleton curve of ECB400 reaches the ultimate load (P_{max}) at about θ = 4.0 %. The degradation in the lateral load of ECB400 is slight at the post-ultimate load stage, and the lateral load is still greater than 0.85 P_{max} at the end of loading ($\theta = 6.5$ %). The slope of skeleton curve of ECB300 at the early loading stage is smaller than that of ECB400, and the peak load of ECB300 is reached at $\theta = 7.5$ %. For ECB300, the lateral load decreases rapidly after the ultimate stage, and the lateral load P drops to about 53 % of P_{max} at the end of loading ($\theta = 8.5$ %).

The equivalent viscous damping coefficient (h_e) is commonly used to evaluate the energy dissipation capacity of structures. According to the Chinese standard JGJ/T 101–2015 [17], h_e can be calculated using Eq. (2) for a certain hysteresis loop:

$$h_e = \frac{1}{2\pi}\frac{S_{ABCD}}{S_{OBE} + S_{ODF}} \tag{2}$$

where S_{ABCD} is the area enclosed by a certain hysteresis loop, and S_{OBE} and S_{ODF} are the areas of triangles OBE and ODF (Fig. 9a).

The relationships between h_e and δ of both specimens are shown in Fig. 9b, from which much better energy dissipation capacity of ECB400 can be found than ECB300. As shown in Fig. 9b, during the early loading stage ($\theta < 2$ %), the h_e values of both specimens decreased progressively as θ increased, which is mainly due to the elastic-dominated deformations of this phase. When θ exceeded 2 %, h_e of ECB300 slightly increased with θ and reached a maximum value of 13.4 % at $\theta = 5$ %, and then gradually decreased to about 9 % at $\theta = 8.2$ %. Compared to ECB300, the increase of h_e of ECB400 is much dramatical at $\theta > 2$ %, and achieved the peak value of 28.8 %, being 2.15 times that of ECB300.

3. Finite element modeling

To comprehensively investigate the effects of main parameters such as column size, embedment depth, axial force ratio and material strength on the hysteresis responses of embedded CFST column base connections, a refined finite element (FE) model is developed employing ABAQUS. Due to the symmetry, only half of each specimen is built in FE modeling.

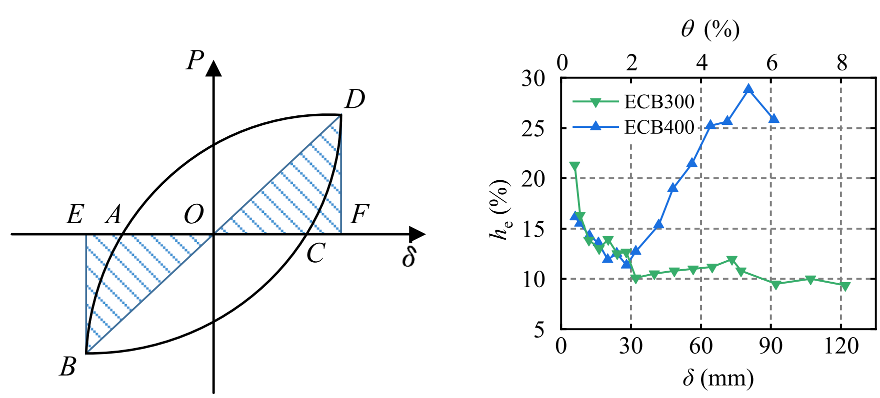
3.1. Mesh and elements

Fig. 10 illustrates the mesh configuration and element types adopted in the FE models. The concrete foundation and infilled concrete are both modeled with 8-node solid elements (C3D8R); the steel tube is represented by 4-node shell elements (S4R), and the steel rebars are simulated using 2-node truss elements (T3D2).

Fig. 11 shows the results of convergence study of FE modeling, where the mesh sizes are changed from 10 mm to 150 mm. It can be found that the differences in ultimate loads are less than 5 % when the element size of FE model of the embedded CFST column base connections is smaller than 50 mm. To balance the accuracy and computational cost, an element mesh size of 50 mm was adopted for the concrete foundation, with a locally refined mesh size of 25 mm in contact areas with other components. The mesh size of 25 mm were used for other components.

3.2. Loading and boundary conditions

In FE modeling, the steel rebars are embedded in the concrete



(a) Calculation method of h_e (b) h_e-δ curve

Fig. 9. Equivalent viscous damping coefficient.

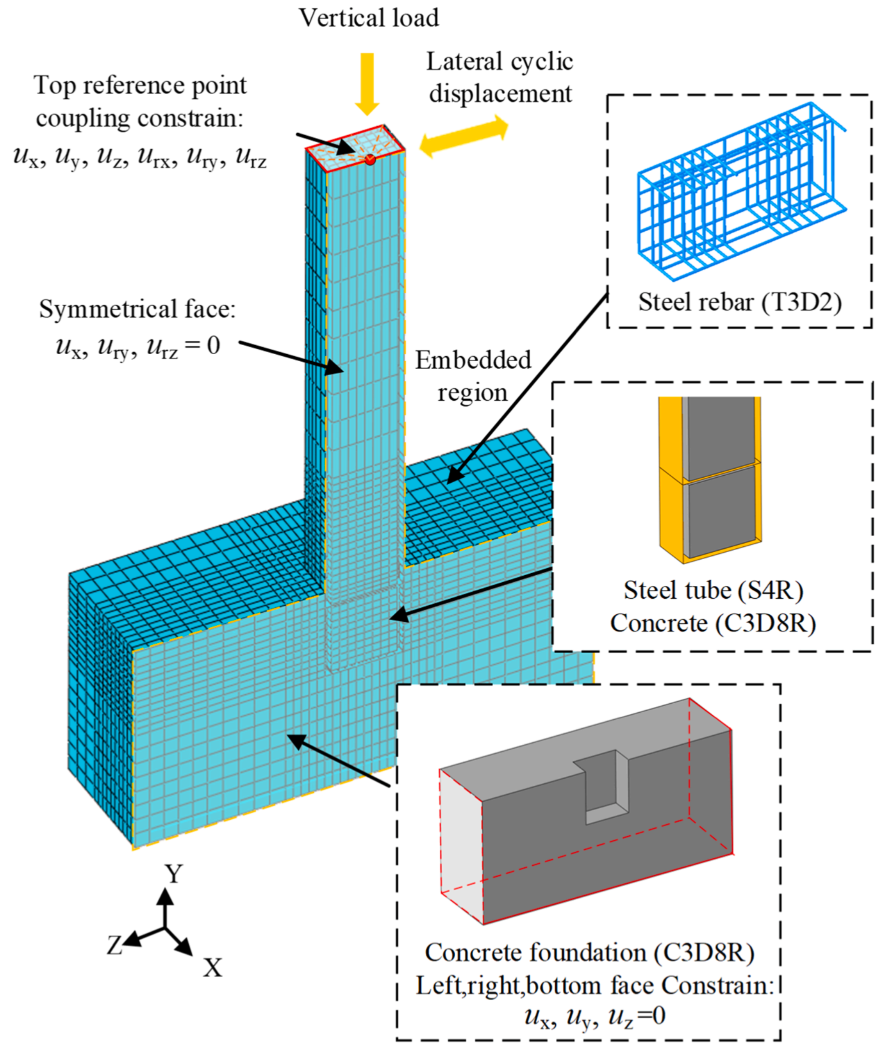


Fig. 10. Details of FE modeling.

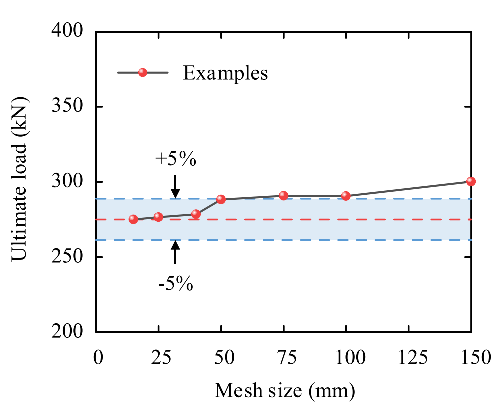


Fig. 11. Mesh sensitivity analysis.

foundation using the "embedded region" constraint. The "surface-to-surface" contacts are defined for modeling the steel-to-concrete interfaces, incorporating the "hard contact" in the normal direction and Coulomb friction law in the tangential direction. Based on previous studies [18], the friction coefficient between steel and concrete surfaces may range from 0.1 to 0.6, and a friction coefficient of $\mu = 0.4$ is adopted according to GB 51022–2015 [19]. A symmetric plane is defined in FE model for applying the symmetry boundary conditions. Additionally, the translation degrees of freedom at the left, right and bottom surfaces of the concrete foundation are restrained. At the top end section of CFST column, a rigid plane is created with all degrees of freedom of nodes coupled (i.e. u_x, u_y, u_z, u_{rx}, u_{ry} and u_{rz}) with respect to the reference point, locating at the center of the top end section (see Fig. 10). The constant compressive load and cyclic lateral load are then applied at the reference point to simulate the experimental loading conditions.

3.3. *Material modeling*

3.3.1. *Modeling of concrete*

The Concrete Damage Plasticity (CDP) model is adopted for simulating concrete material. The constitutive model specified in GB 50010–2010 [20] for the plain concrete is used to simulate the compressive behavior of the concrete foundation, as shown in Eqs. (3) and (4)).

$$d_c = \begin{cases} 1 - \dfrac{\rho_c n}{n - 1 + x^n} x \le 1 \\ 1 - \dfrac{\rho_c}{\alpha_c (x-1)^2 + x} x > 1 \end{cases} \tag{3}$$

$$\sigma_c = (1 - d_c) E_c \varepsilon_c \tag{4}$$

where the parameters ρ_c and n can be taken as $f'_c / E_c \varepsilon_{c,r}$ and $E_c \varepsilon_{c,r} / (E_c \varepsilon_{c,r} - f'_c)$, respectively; f'_c is the cylindrical concrete compressive strength; $\varepsilon_{c,r}$

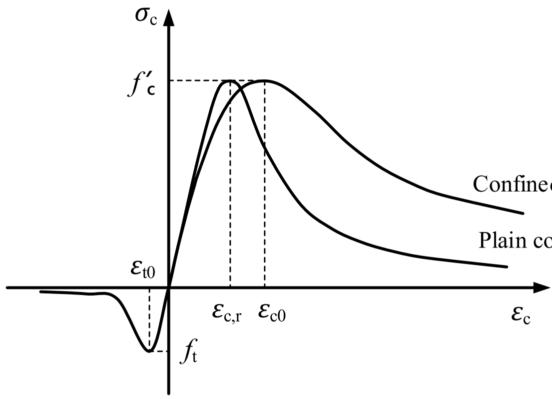


Fig. 12. Constitutive models of concrete used in ABAQUS.

is the strain corresponding tof'_c; α_c is a parameter reverent to the descending branch of the stress-strain curve; and E_c is the elastic modulus of concrete, taken as $4700\sqrt{f'_c}$ according to the Eurocode 4 [21].

A number of confined concrete constitutive models have been proposed in literature (e.g. Mander [22] and Han *et al.* [23]), and the model proposed by Han *et al.* is used to describe the compressive behavior of the concrete filled in the CFST column this study (Fig. 12), as given Eqs. (5 and 6).

$$\frac{\sigma_c}{f'_c} = \begin{cases} 2\cdot\varepsilon_c/\varepsilon_{c0} - (\varepsilon_c/\varepsilon_{c0})^2 & \varepsilon_c \le \varepsilon_{c0} \\ \dfrac{\varepsilon_c/\varepsilon_{c0}}{\beta_0\cdot(\varepsilon_c/\varepsilon_{c0}-1)^{1.6+1.5\varepsilon_c/\varepsilon_{c0}} + \varepsilon_c/\varepsilon_{c0}} & \varepsilon_c > \varepsilon_{c0} \end{cases} \tag{5}$$

$$\xi = \frac{A_s f_y}{A_c f_{ck}} \tag{6}$$

where ε_{c0} is the strain corresponding to the peak stress of the confined concrete; ξ is the confinement factor; A_s and A_c are the cross-sectional areas of steel and concrete, respectively; and f_{ck} is the characteristic compressive stress of concrete and can be taken as 0.67 f_{cu}.

For modeling the concrete in tension, the stress-strain relationship proposed by Shen *et al.* [24] is adopted:

$$\frac{\sigma_t}{f_t} = \frac{\varepsilon_t/\varepsilon_{t0}}{0.31 f_t^2(\varepsilon_t/\varepsilon_{t0})^{1.7} + \varepsilon_t/\varepsilon_{t0}} \tag{7}$$

$$f_t = 0.26 \times (1.25 f'_c)^{2/3} \tag{8}$$

in which ε_{t0} is the strain corresponding to the peak tensile stress, and f_t is the tensile strength of the concrete.

The focal point model [25] is employed to simulate the concrete damage under cyclic loading, in which the parameters for the concrete damage under compression and tension, d_c and d_t, can be given by:

$$d_c(\sigma_c, \varepsilon_c) = 1 - \frac{\sigma_c + n_c\sigma_{c0}}{E_c(\varepsilon_c + n_c\sigma_{c0}/E_c)} \tag{9}$$

$$d_t(\sigma_t, \varepsilon_t) = 1 - \frac{\sigma_t + n_t\sigma_{t0}}{E_c(\varepsilon_t + n_t\sigma_{t0}/E_c)} \tag{10}$$

in which n_c and n_t are the compressive and tensile damage variations coefficients, which are taken as $n_c = 2.0$ and $n_t = 1.0$ for the rectangular concrete-filled steel tubular column [26], respectively.

3.3.2. Modeling of steel

The mixed hardening model, based on the von Mises yield criterion and the Prandtl-Reuss flow rule, is employed in the present study to simulate the behavior of steel under cyclic loading. As illustrated in Table 1, the key parameters of the mixed hardening model are determined according to the study of Shi *et al.* [27,28], where C_i and γ_i are the parameters to describe the plastic flow of the yield surface.

Table 1
Parameters of the mixed hardening model of steel.

Q_∞ (MPa)	b	C_1 (MPa)	γ_1	C_2 (MPa)	γ_2	C_3 (MPa)	γ_3
21	1.2	6013	173	5024	120	3026	32

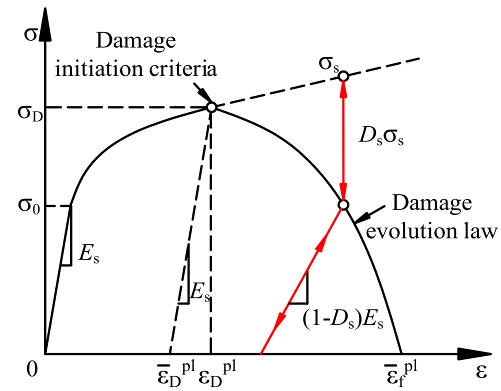


Fig. 13. Ductile damage stress-strain constitutive model of steel.

The ductile damage model is implemented to characterize the progressive strength degradation and stiffness deterioration of structural steel under cyclic loading due to the accumulated damage, where the damage behavior is described using the damage initiation criteria and the damage evolution laws (Fig. 13). The damage index D_s can be given as [29]:

$$\overline{E}_s = (1 - D_s)E_s \tag{11}$$

$$\overline{\sigma}_s = (1 - D_s)\sigma_s \tag{12}$$

$$D_s = 0.96(\overline{u}^{pl}/\overline{u}_f^{pl})^{2.42} \tag{13}$$

where $\overline{E}_s$ and $\overline{\sigma}_s$ are the elastic modulus and stress with damage; $\overline{u}^{pl}$ and $\overline{u}_f^{pl}$ are the plastic displacement and plastic damage displacement (Eq. (14)) [30]. For the case of $\overline{u}^{pl} = 0$, the damage occurs (D_s=0) with E_s and σ_s neither reduced, and steel is completely damaged (D_s=1) at $\overline{u}^{pl} = \overline{u}_f^{pl}$, with both the elastic modulus and stress becoming zero.

$$\overline{u}_f^{pl} = L_{char}(\overline{\varepsilon}_f^{pl} - \overline{\varepsilon}_D^{pl}) \tag{14}$$

where L_{char} is the element size of steel tube, taken as 25 mm in this study.

It is worth mentioning that the ductile damage model is believed to be beneficial for accurately simulating the strength degradation and stiffness deterioration of steel under cyclic loading when the local buckling of CFST column is involved. A similar modeling approach was adopted in other studies (e.g. [31]).

3.4. Validation of FE modeling

In validating FE modelling, the Young's modulus and strength of steel and concrete are taken from the coupon tests (E_s = 193.7 GPa, E_c = 25.3 GPa, f_y = 279.8 MPa and f_{cu} = 36.8 MPa). As shown in Fig. 7, the hysteresis curves from FE analyses and tests are in good agreements, even for the pinching responses in the second and fourth quadrants and the sliding phenomenon near the *x*-axis of hysteresis curves. As illustrated in Table 2, the ultimate loads from the test and FE analyses, P_{Test}

Table 2
Comparison between test and FE results.

Specimens	Loading Direction	P_{Test} (kN)	P_{FEM} (kN)	P_{Test}/P_{FEM}
ECB400	(+)	231.7	239.4	0.97
	(-)	254.4	239.4	1.06
ECB300	(+)	227.1	211.2	1.08
	(-)	234.0	209.2	1.12

and P_{FEM}, of both specimens agree very well.

The failure modes of both specimens from experimental tests and FE analyses are compared in Fig. 14. As shown in Figs. 14a and 14b, the concrete crack patterns from FE analyses, quantified through DAMAGET (tensile damage parameter) contours. It can be seen from Fig. 14. that concrete cracking develops around the CFST column for ECB300, whereas the concrete cracks are mainly concentrated at the corners of the CFST column for ECB400, demonstrating a strong correlation with experimental observations. The other main features, such as the separation between the CFST column and surrounding concrete of ECB300 and local buckling of outer steel tube of ECB400, are also well captured in FE analyses (i.e. Figs. 14c and 14d). Moreover, it can be seen from Figs. 7a and 7b that very good agreements can be found in the comparisons of hysteresis responses of both specimens from test and FE analyses. The above comparisons indicate that the FE modeling presented in this study is capable of simulating the hysteresis behaviors of the embedded CFST column base connections, and thus the presented FE modeling will be adopted in the following parametric study.

4. Parametric analysis

A systematic parametric study is conducted to evaluate the influences of main parameters, such as the width of CFST cross-section (B), thickness of steel tube (t), embedment depth (d_{embed}), axial load ratio (n), strength of concrete foundation (f_{cu}), strength of filled concrete ($f_{cu,fill}$), concrete strength ratio ($\alpha_c = f_{cu}/f_{cu,fill}$) and width of concrete foundation (W_b) on the performance of embedded CFST column base connections. For simplification, the width B (perpendicular to the loading direction) and the height H (parallel to the loading direction) of CFST sections are assumed to be the same.

Table 3 lists all numerical models established in FE analyses. Three cross-sections of CFST column with small width-to-thickness ratios (i.e. B250 ×250 ×20(B250), B400 × 400 × 40(B400) and B550 × 550 × 55 (B550)) are adopted in parametric study, which is intended to prevent the premature local buckling of outer steel tubes. To avoid adverse effects of the insufficient distances between the CFST column and free edges of concrete foundation, the ratio of W_b/B should be not less than 3.0 [16], and thus the dimensions of concrete foundations for B250, B400 and B550 examples are designed as (length × width × height), 1460 × 820 × 700 (mm), 1752 × 1230 × 1050 (mm), and 2190 × 1640 × 1050 (mm), respectively. The loading point to the top surface of the concrete foundation is taken as 1500 mm for all numerical examples, representing one half of the typical story height (i.e. 3000 mm).

The yield strength (f_y) and elastic modulus (E_s) for material of steel tubes and steel rebars are set to 235 MPa and 206000 MPa, respectively. The Poisson's ratios (ν) of concrete and steel are taken as 0.2 and 0.3, respectively. The applied axial load (N) is represented using the axial load ratio (n).

$$n = \frac{N}{N_{sc}} \tag{15}$$

$$N_{sc} = f_y A_s + 0.85 f'_c A_c \tag{16}$$

where N_{sc} is the compressive resistance of the composite section [21]; and A_s and A_c are the areas of steel tube and infilled concrete.

4.1. Effect of embedment depth

Fig. 15 compares the skeleton curves of numerical examples with different embedment depths (d_{embed}). As shown in Fig. 15, the ultimate bending moments (M_u) increase with the embedment depth for examples with both B250 and B400 sections. With increasing d_{embed}, the lateral loads gradually level off at the post-ultimate stage, indicating an

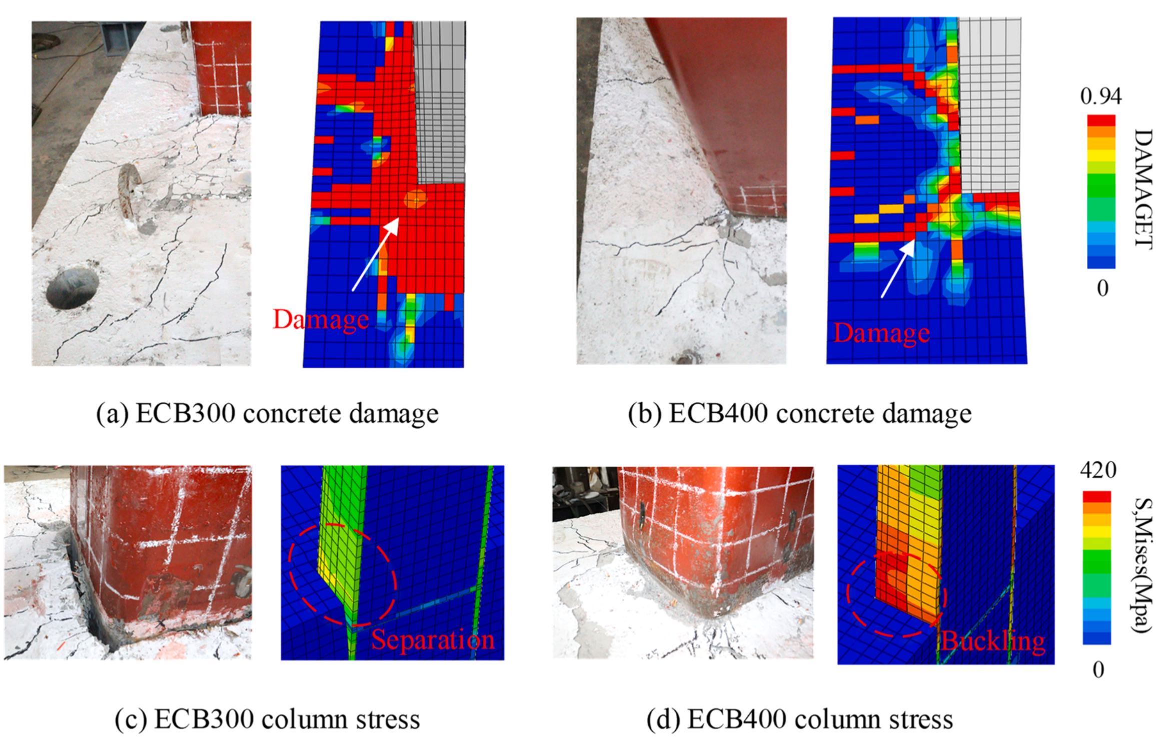


Fig. 14. Failure modes of tests and FE models.

Table 3
Details of specimens in parametric study.

Examples	d_{embed} (mm)	n	t (mm)	$B{=}H$ (mm)	f_{cu} (MPa)	$f_{cu,fill}$ (MPa)	W_b (mm)
B-250-d−100	100	0	20	250	35	35	820
B-250-d−200	200						
B-250-d−300	300						
B-250-d−400	400						
B-250-d−500	500						
B-400-d−200	200	0	40	400	35	35	1230
B-400-d−400	400						
B-400-d−600	600						
B-400-d−800	800						
B-250-n−0	300	0	20	250	35	35	820
B-250-n−0.2		0.2					
B-250-n−0.4		0.4					
B-400-n−0	400	0	40	400	35	35	1230
B-400-n−0.2		0.2					
B-400-n−0.4		0.4					
B-550-n−0	550	0	55	550	35	35	1640
B-550-n−0.2		0.2					
B-550-n−0.4		0.4					
B-250-f_{cu}−35	300	0	20	250	35	35	820
B-250-f_{cu}−50					50		
B-250-f_{cu}−60					60		
B-400-f_{cu}−35	300	0	40	400	35	35	1230
B-400-f_{cu}−50					50		
B-400-f_{cu}−60					60		
B-550-f_{cu}−35	300	0	55	550	35	35	1640
B-550-f_{cu}−50					50		
B-550-f_{cu}−60					60		
B-250-α_c−0.6	300	0	20	250	35	60	820
B-250-α_c−0.7						50	
B-250-α_c−1.0						35	
B-400-α_c−0.6	300	0	40	400	35	60	1230
B-400-α_c−0.7						50	
B-400-α_c−1.0						35	
B-550-α_c−0.6	300	0	55	550	35	60	1640
B-550-α_c−0.7						50	
B-550-α_c−1.0						35	
B-250-W_b−620	300	0	20	250	35	35	620
B-250-W_b−820							820
B-250-W_b−1020							1020
B-250-W_b−1220							1220
B-250-W_b−1420							1420
B-400-W_b−820	300	0	40	400	35	35	820
B-400-W_b−1020							1020
B-400-W_b−1220							1220
B-400-W_b−1620							1620
B-400-W_b−2220							2220

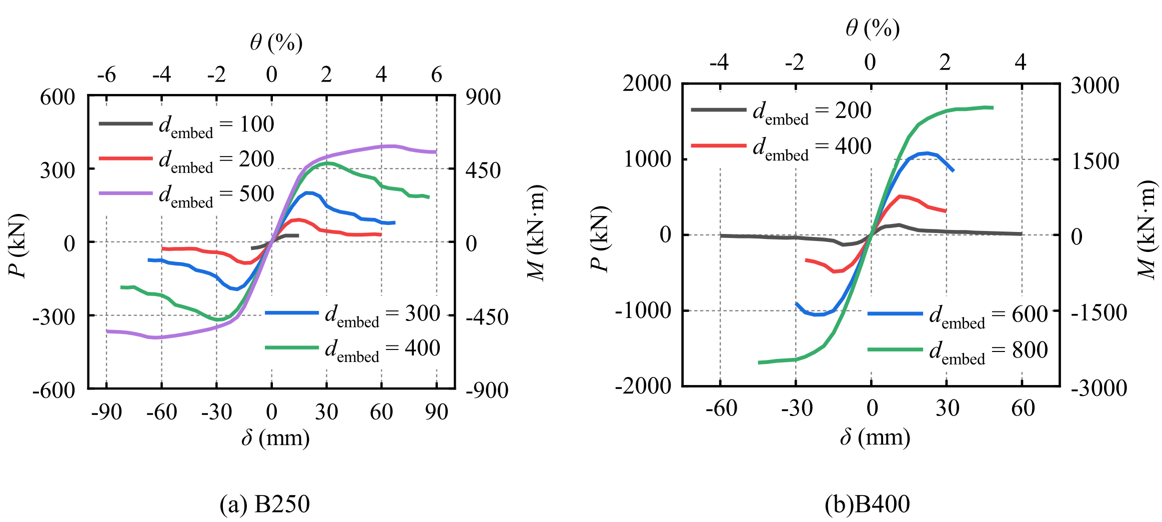


Fig. 15. Effect of embedment depth on skeleton curves.

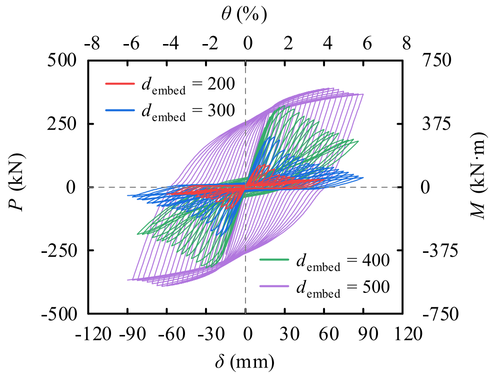


Fig. 16. Effect of embedment depth on hysteresis curves (B250).

improvement in ductility. Additionally, the drift ratio (θ) corresponding to M_u also increases with a greater embedment depth.

It should be noted that besides the improvements in the ultimate load and ductility due to the increase of embedment depth shown in Fig. 15, the hysteresis behaviors of the examples are also enhanced (see Fig. 16). It can be seen from Fig. 16 that the hysteretic responses of B250 examples with the embedment depths being 200 mm, 300 mm, and 400 mm, the hysteresis curves exhibit the pronounced pinched responses, caused by the concrete crushing around CFST section. In contrast, for the case of $d_{embed} = 500$ mm, the hysteresis curve becomes more rounded, primarily due to the plastic deformation of the steel tube.

It can be seen from Fig. 17 that, for examples with B250 section, the concrete damage (respected using DAMAGEC) occurs throughout the entire embedment region when the embedment depths are relatively small ($d_{embed} \leq 300$ mm). However, as the embedment depth increases ($d_{embed} > 300$ mm), the damage of concrete becomes concentrated at the upper and lower regions of the embedment depth. For the case of $d_{embed} = 500$ mm, the failure mode shifts from concrete bearing failure to steel tube buckling. The contour figure shown in Fig. 17e indicates that the Mises stress in the steel tube may exceed the yield strength (f_y) of steel and enter the hardening stage. The damage behaviors of examples with B400 section are very similar to those with B250 section, which are not given in this study for brevity.

4.2. Effect of axial load ratio

Comparisons shown in Fig. 18 between the results of examples without axial load ($n = 0$) and those of examples with axial loads ($n = 0.2$ and 0.4) indicate that applying the axial load may improve the ultimate resistance of column base connections. The ultimate bending moments (M_u) of $n = 0.2$ are about 1.4, 2.3, and 3.0 times those of $n = 0$ for examples with B250, B400 and B550 sections, respectively. For examples with B250 section (Fig. 18a), it is noteworthy that the M_u for the cases of $n = 0.2$ and $n = 0.4$ are relatively close, as their behaviors are governed by the flexural strength of CFST section rather than the crushing of concrete foundation.

Fig. 19 shows the vertical stress distributions in the concrete foundation below the bottom end of CFST columns at the ultimate stages (M_u), to illustrate the reaction stress from the concrete foundation to column bottom end. It can be seen from Fig. 19 that only a portion of cross-section of column bottom end is compressed by the reaction stress at the ultimate stage. Comparing the cases with $n = 0.2$ and $n = 0.4$ shown in Fig. 19 implies that the compressive regions blow the column end may increase due to the increase of axial force, and the widths of compressive areas along the loading direction for all examples are not greater than $H/2$. This provides a basis for considering the effect of the axial force on the ultimate load of the embedded column base connections.

4.3. Effect of material strength

The load-displacement relationships of examples with different concrete strengths, f_{cu}, are plotted in Fig. 20, which indicate that both the stiffness and ultimate moment (M_u) increase as f_{cu} increases. Note that the drift ratio (θ) corresponding to M_u remains nearly constant regardless of the concrete strength for all cases.

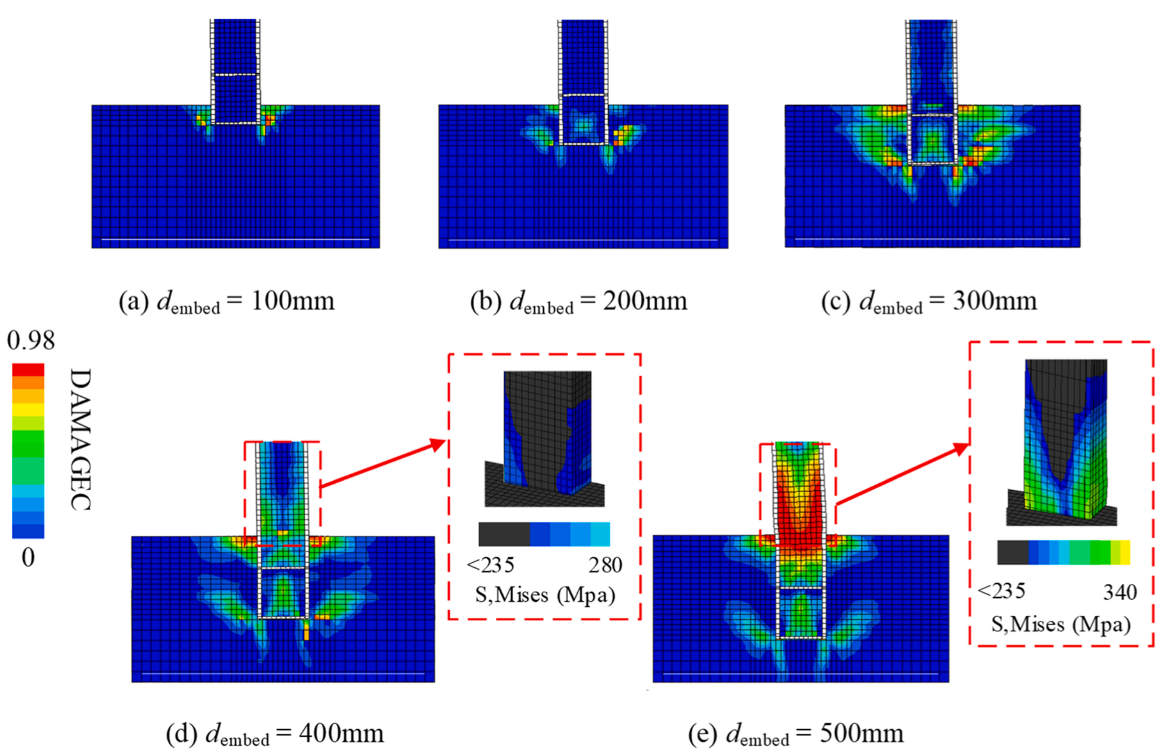
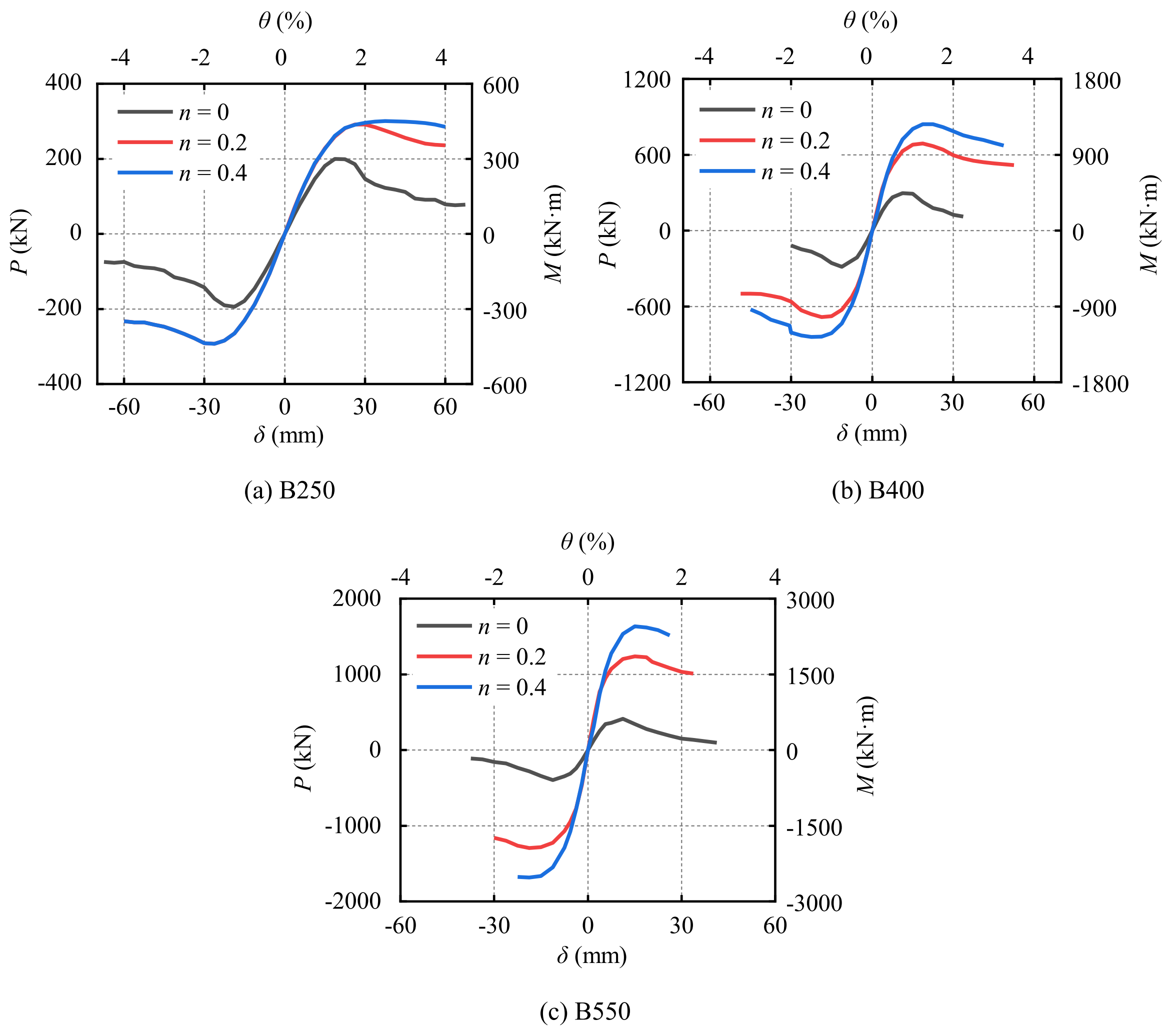


Fig. 17. Concrete damage and Mises stress contours at failure (B250).



(a) B250 (b) B400

(c) B550

Fig. 18. Effect of axial load ratio on skeleton curves.

In practical engineering applications, the strength of concrete filled in CFST columns may be higher than that of foundations. Therefore, the effects of concrete strength ratio ($\alpha_c = f_{cu}/f_{cu,fill}$) on the behavior of embedded column base connections are also analyzed in this section. As shown in Fig. 21, for all numerical examples, the effects of concrete strength ratio are very slight and may be negligible for column base connections with different CFST sections (i.e. B250, B400 and B550). This is because the behavior of the column base connection is mainly related to the concrete foundation when the concrete foundation failure governs the strength of column base connection due to the insufficient embedment.

4.4. Effect of foundation width

The effects of W_b on behavior of the embedded CFST column base connection are shown in Fig. 22. It can be seen from the comparisons of skeleton curves of examples with B250 and B400 sections that the ultimate load of the embedded CFST column base connections may increase by using a wide concrete foundation, while for the width ratio of the concrete foundation to CFST column greater than about 3.0 (3.3 for B250, 3.05 for B400), further increase in W_b does not bring the obvious improvement of the ultimate load. This also matches the provisions of the code [16].

5. Ultimate load models

5.1. Possible ultimate states

The strength of embedded CFST column base connections may be dominated by three possible modes (see Fig. 23): the bearing failure of surrounding concrete, the CFST column failure and upward punching failure of concrete foundation. The ultimate moments corresponding to these three failure modes are denoted as M_c, M_y and M_{pun}, respectively. Then, the ultimate load of CFST embedded column base (M_{The}) can be determined as:

$$M_{The} = \min(M_c, M_y, M_{pun}) \tag{17}$$

As M_y can be calculated based on provisions of the Eurocode 4 [21], the analytical models for calculating M_c and M_{pun} are only discussed in this study.

5.2. Concrete Bearing failure

When the column's embedment depth is insufficient, the strength of column base connections is frequently controlled by concrete bearing failure. According to existing studies (e.g. [10] and [32]), the bearing stress in surrounding concrete of the embedded steel column base connection may degrade along the depth of embedment, which indicates that only a certain depth may be only effective. Therefore, an effective embedment depth, d_{eff}, is defined in this study for the bearing stress

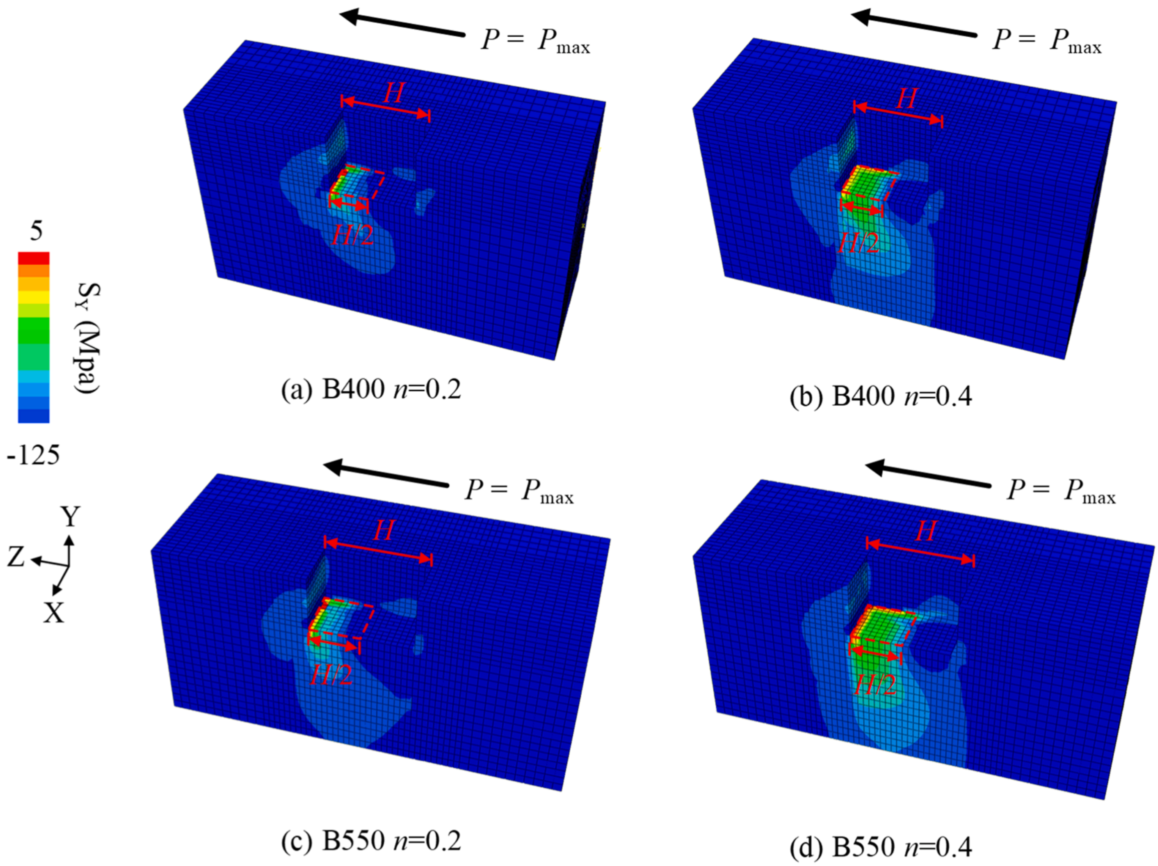


Fig. 19. Stress distributions below the bottom end of CFST column.

analysis. Based on the solution of Hetenyi [32], which was on an end-loaded beam embedded in a semi-infinite elastic medium, Grilli *et al.* [10]derived the following expression of d_{eff} for H-section steel embedded column bases:

$$d_{\mathrm{eff}} = \frac{C_0}{\rho} \leq d_{\mathrm{embed}} \tag{18}$$

$$\rho = \left[\frac{E_c}{4(EI)_{\mathrm{section}}}\right]^{1/4} \tag{19}$$

in which E_c are the Young's modulus of the concrete base; $(EI)_{\mathrm{section}}$ is the bending rigidity of the CFST column; d_{embed} is the embedment depth of column base; and C_0 is a constant and can be taken as $\pi/2$ according to Xu *et.al* [12].

For ease of description, all external forces and resultant sectional forces are calculated with respect to the reference point (see Fig. 24, denoted as reference section hereafter), which is the centroid of the column section at the top surface of concrete foundation. As shown in Fig. 24, the external loads result in the axial load N, shear force V, and bending moment M of the reference section. The assumed distribution pattern of the bearing stresses in the surrounding concrete, induced by the interaction between the CFST column and the surrounding concrete, is also shown in Fig. 24. The force equilibrium in the horizontal direction gives

$$f_{\mathrm{bearing}} B(d_t - d_b) = V + F_f \tag{20}$$

$$d_t + d_b = d_{\mathrm{eff}} \tag{21}$$

where d_t and d_b are the top and bottom heights of bearing stress blocks; F_f is the friction between the column bottom end and concrete foundation; and f_{bearing} is the ultimate compression stress of concrete.

The f_{bearing} is based on the local bearing strength as specified in GB 50010–2010 [20] and Eurocode 2 [33], and can be expressed as follows:

$$f_{\mathrm{bearing}} = 1.35\beta_c\beta_l f_{\mathrm{ck}} \tag{22}$$

$$\beta_l = \sqrt{\frac{A_b}{A_l}} \tag{23}$$

where β_c is a reduction factor of concrete strength, and can be taken as 1.0 for $f_{\mathrm{cu}} < 50$ MPa; A_b is the loading area; and A_l is the design area in the similar shape of A_b. According to [20], for the considered case, the strength enhancement factor for the localized compression (β_l) can be taken as 1.7.

The friction force, F_f, at the bottom end of the column is due to the axial compressive force (N), where the friction coefficient μ is taken as 0.4 according to GB 51022–2015 [19].

$$F_f = \mu N \tag{24}$$

As illustrated in Fig. 24, the force and moment balances lead to:

$$V + F_f = f_{\mathrm{bearing}} B(d_t - d_b) \tag{25}$$

$$N = f_{\mathrm{bottom}} BH/2 \tag{26}$$

$$M + 0.5 f_{\mathrm{bearing}} B d_t^2 = F_f d_{\mathrm{embed}} + f_{\mathrm{bearing}} B d_b \left(d_t + \frac{d_b}{2}\right) + N\frac{H}{4} \tag{27}$$

It should be noted that the compressive width of axial force along the loading direction at the ultimate state is assumed to be $H/2$ in Eq. (26), based on the discussion in previous section (see Fig. 19). The ultimate shear force and moment, V_u and M_u, can be given by Eqs. (28) and (29), where the relationships $d_t = d_{\mathrm{eff}}$ and $d_t = d_b = 0.5d_{\mathrm{eff}}$ are adopted, respectively. Note that f_{bottom} and F_f both become zero when the axial

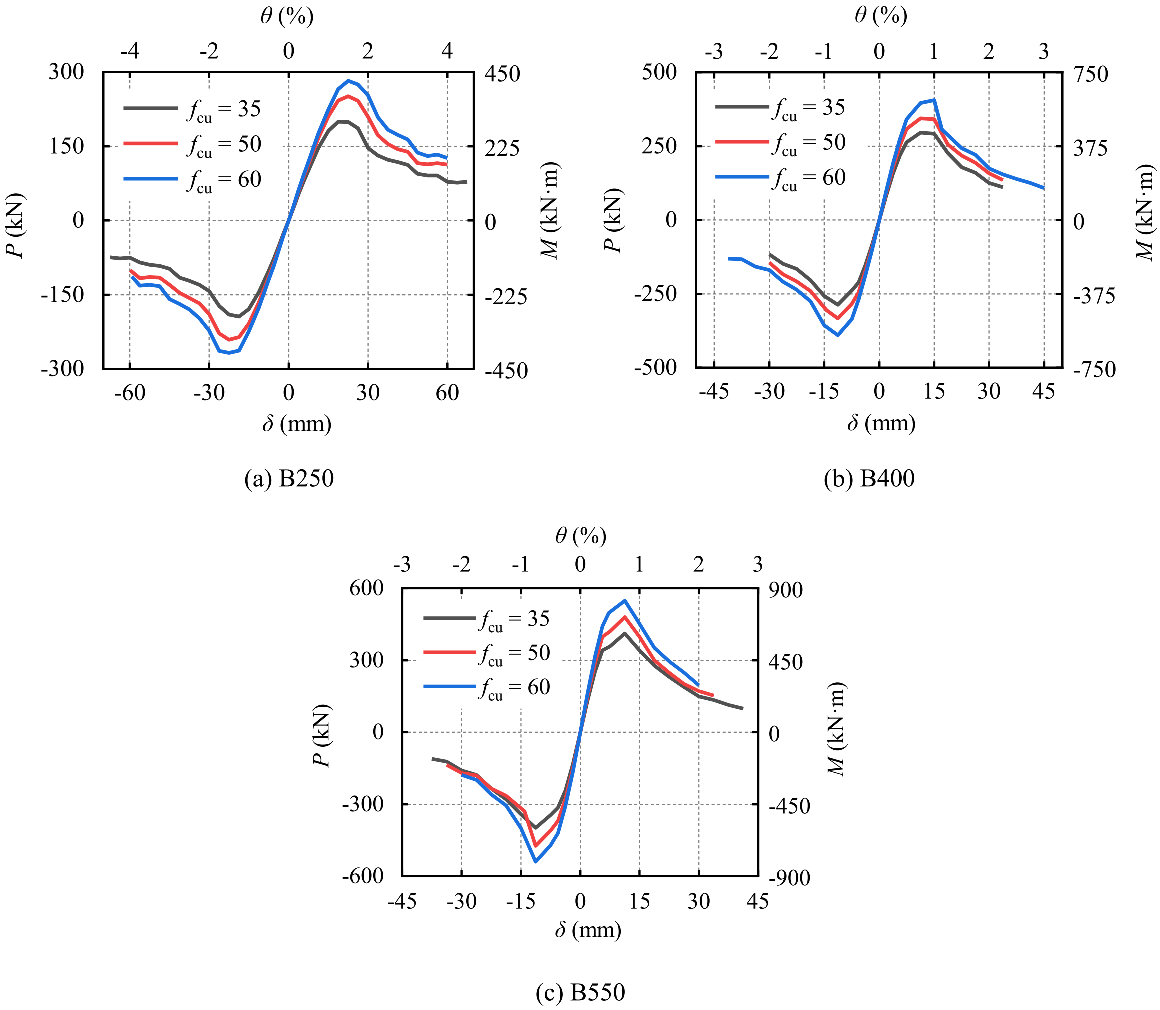


Fig. 20. Effect of concrete strength on skeleton curves.

load N is not considered.

$$V_u = f_{bearing} B d_{eff} \tag{28}$$

$$M_u = f_{bearing} B \left(0.25 d_{eff}^2\right) = 0.25 V_u d_{eff} \tag{29}$$

Based on Eqs. (25–27) and simplified using Eqs. (28) and (29), the V-M relationship with the axial load N taken into account can be given by:

$$\left(\frac{V + \mu N}{V_u}\right)^2 + \frac{M + 0.5 V d_{eff} - N(0.25H + \mu d_{embed} - 0.5 \mu d_{eff})}{M_u} = 1 \tag{30}$$

To provide a more intuitive view of the influence of axial force on the concrete bearing capacity of embedded column base, the normalized V-M relationships of B250 examples under different axial loads are plotted in Fig. 25, where all results are calculated using Eq. (30). The comparisons in Fig. 25 indicates that, as far as the concrete bearing strength is considered, the axial force is beneficial to load-carrying capacity of the embedded CFST column base.

5.3. Upward punching failure

When the embedment depth (d_{embed}) of CFST column is small or the reinforcement of vertical rebars is insufficient, the concrete foundation may fail by the upward punching shear failure (Fig. 26). Similar to the concrete bearing failure analysis, it is assumed that only one half of the bottom end section of CFST column is under compression (i.e. $H/2$) at the ultimate state (Fig. 19). The upward punching shear force (F_{pun}), by accounting for the contributions of concrete and stirrups and assuming a 45° punching shear failure surface [20], can be expressed as follows:

$$F_{pun} = 0.7 \beta_h f_t S + 0.8 f_{yv} n_s A_s \tag{31}$$

where β_h is the height influence factor and can be taken as 1.0 when the height of the punching shear face≤800 mm; S is the projected area of the punching shear failure face; f_{yv} is the yield strength of the stirrups; A_s is the cross-sectional area of single stirrup; and n_s is the number of stirrups intersecting the punching shear face.

According to the force balance condition in the horizontal direction, the height of the concrete bearing area, x, can be expressed as:

$$x = \frac{V + \mu(N + F_{pun})}{f_{bearing} B} \tag{32}$$

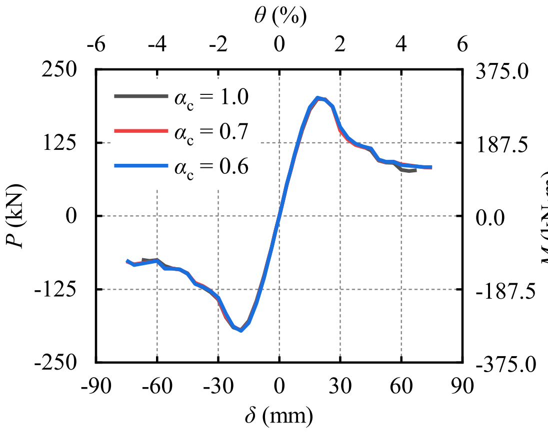
where the term $\mu(N + F_{pun})$ is the friction force acted at the bottom end of CFST column, and $N + F_{pun}$ is the reaction compressive force at the column bottom.

The equilibrium condition of the bending moment with respect to the reference section gives:

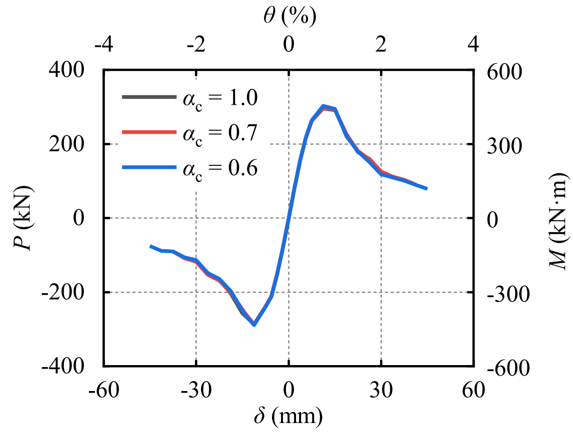
$$M_{pun} = F_{pun}[(0.5 + \mu) d_{embed} + 3H/4] + N(\mu d_{embed} + H/4) - f_{bearing} B x^2 \big/ 2 \tag{33}$$

5.4. Comparison of predictions

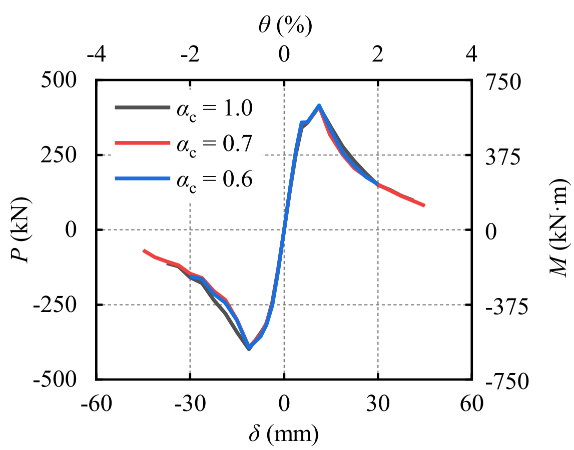
The results of the above mentioned failure modes for all examples are



(a) B250

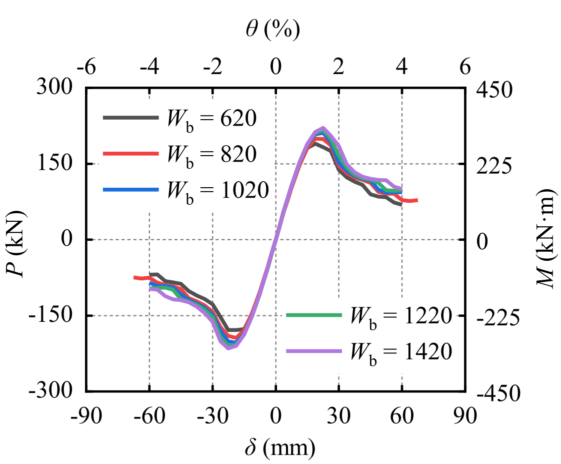


(b) B400

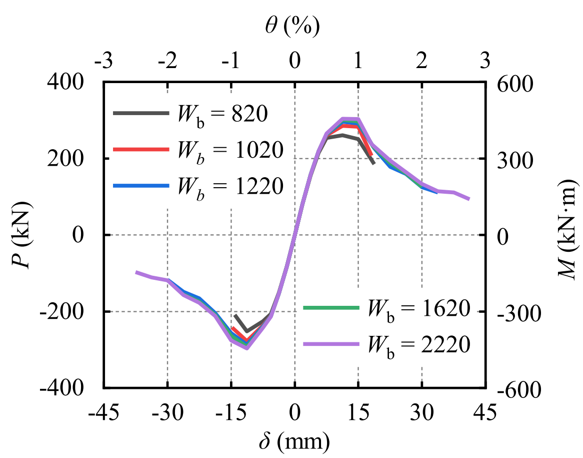


(c) B550

Fig. 21. Effect of the strength ratio of foundation and filled concrete.



(a) B250



(b)B400

Fig. 22. Effect of base width on skeleton curves.

listed in Table 4. As illustrated in Table 4, the strengths of most examples are governed by the concrete bearing failure (CB), especially for those with the relatively shallow embedment. For examples with relatively deep embedment, the section failure (SS) of CFST column may be the dominant failure mode. The UPS failure may become the controlling failure mode for examples with the high concrete strength incorporating the relatively shallow embedment, as illustrated in Table 4.

Comparisons are made between the predictions of existing solutions

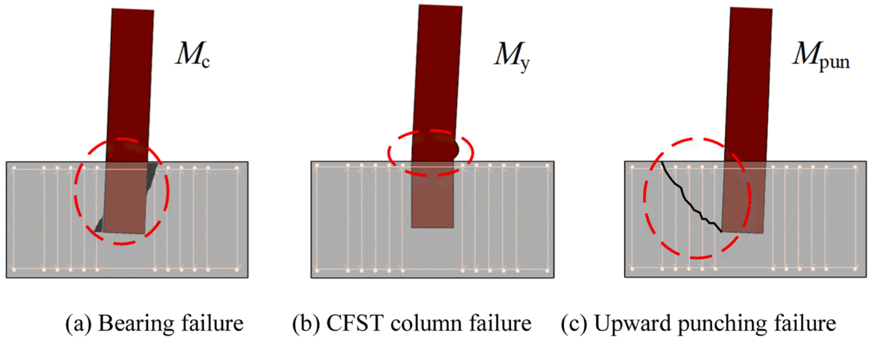


Fig. 23. Failure modes of CFST embedded column base.

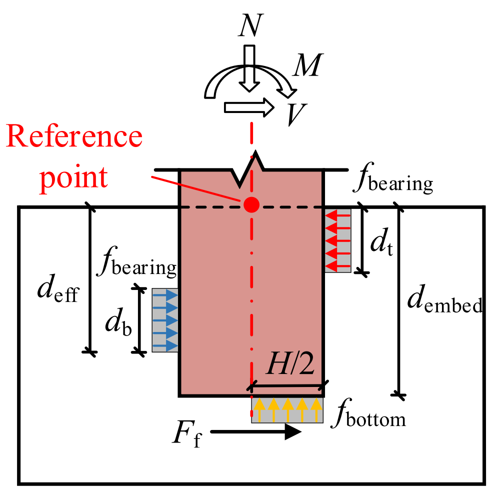


Fig. 24. Stress distribution in concrete base.

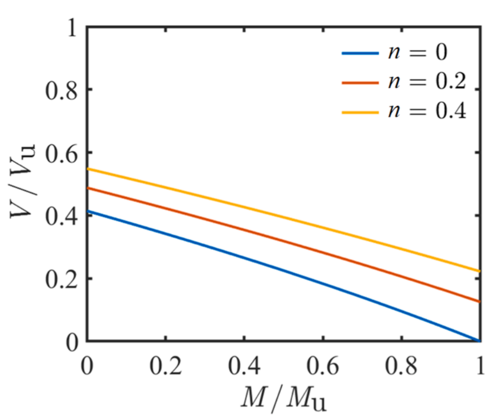


Fig. 25. Normalized *M-V* relationship under different axial loads (B250, d_{embed}=300 mm).

with FE results (M_{FEM}) in Table 5, where M_{AISC}, M_{GB}, M_{JGJ} and M_{Grilli} are respectively using the methods of AISC 341–16 [14], GB 50017–2017 [13], JGJ 138–2016 [15] and Grilli *et al.* [10]. It should be noted that the methods of AISC 341–16 [14], GB 50017–2017 [13], JGJ 138–2016 [15] and Grilli *et al.* [10] are not applicable to the cases with axial loads, and thus the axial loads are not accounted for in their predictions in Table 5.

As illustrated in Table 5, the predictions of the proposed method (M_{The}) match very well with the FE results, with an average ratios (Avg) being 0.95 and a coefficient of variation (CoV) being 0.07. Among all examples, the Avg ratios of the presented method for the examples failed by the concrete bearing failure and upward punching shear failure are 0.94 and 1.02, respectively. It should be noted that the M_{The} marked with "*" are much smaller than M_{FEM}, which is due to the under-estimated predictions of M_y using the code method [21].

The predictions of AISC 341 method [14] are also in good agreements with FE results considering the average ratio (i.e. the Avg being 0.96), while the CoV coefficient of AISC 341 predictions (0.27) is much greater than that of M_{The} (0.07), indicating much higher variations in the performance of M_{AISC}. Particularly, the AISC method tends to over-estimating the ultimate loads of column base connections with a great width of concrete foundation (W_b). This is because the AISC method is based on experimental results of steel sections embedded in concrete shear walls [34,35], where the monotonic increasing relationship between the concrete base width and column width (i.e. $(W_b/B)^{0.66}$, for $W_b/B \leq 12$) is adopted. This is obviously not the case, according to the FE results shown in Fig. 22.

The predictions of other three models, M_{GB}, M_{JGJ} and M_{Grilli}, are generally conservative compared to the FE results, with the Avg coefficients being 0.39, 0.28 and 0.65, respectively. Note that the Grilli's model [10] was presented for the embedded column bases with the extended bottom end plate, and thus the results in Table 5 are obtained by omitting the contribution of extended end plate.

Fig. 27 shows the comparisons between the predictions of the proposed solution and FE results, which indicates very good performance of the proposed solution for the ultimate loads of the embedded column bases.

The moment–rotation relationships are obtained by extracting the rotational angle and moment of the CFST column section at the top surface plane of concrete foundation. In this study, the rotational stiffnesses of column base connections listed Table 4 are determined as the slope of the skeleton curves between −0.005 rad and + 0.005 rad [36].

According to Eurocode 3 [37], the column base can be regarded as rigid connection for moment frames when $K_c \geq 30$, where the dimensionless rotational stiffness K_c can then be defined base on the Eurocode 3 [37] as:

$$K_c = k_c L_s/(EI)_{section} \tag{34}$$

where L_s is the story height of construction and taken as 3000 mm, and k_c is the initial stiffness.

As indicated in the results of Table 4, all analyzed examples do not satisfy the column base criterion for the rigid connection of Eurocode 3 [37]. This is because the relatively shallow embedment depths are adopted for these examples, as the behavior of column base connections controlled by the concrete foundation failure is mainly concerned in this study.

5.5. Comparison of required embedment depth

In engineering application, the embedment depth of the embedded CFST column base connection is essential for the in-site installation. In

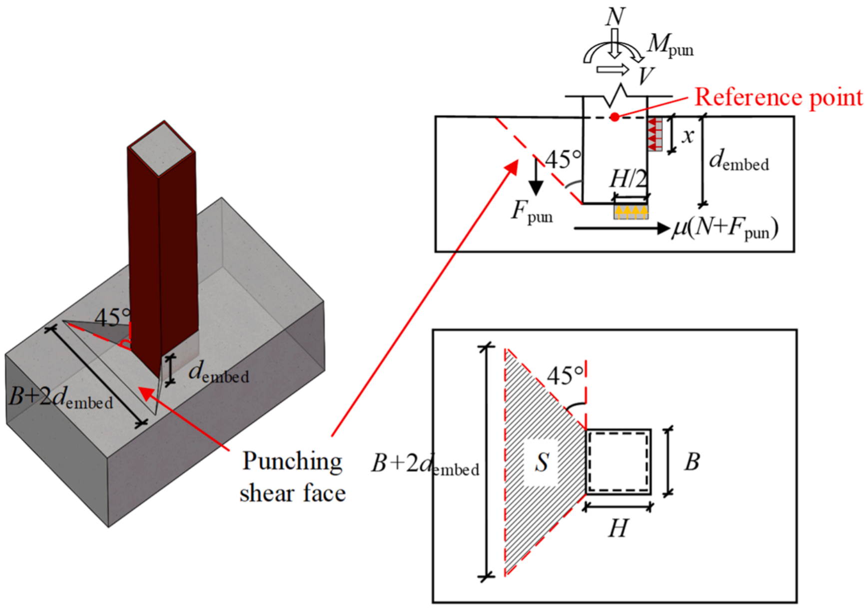


Fig. 26. Mechanism of upward punching shear failure.

Table 4
Ultimate loads and rotational stiffnesses.

Specimens	n_s	M_c (kN·m)	M_y (kN·m)	M_{pun} (kN·m)	M_{The} (kN·m)	Governing Mode	k_c (kN·m/rad)	K_c
B-250-d–100	2	32.5	402.4	51.1	32.5	CB	8363.7	0.67
B-250-d–200	6	125.8	402.4	189.9	125.8	CB	30099.0	2.40
B-250-d–300	10	274.1	402.4	401.0	274.1	CB	57720.6	4.60
B-250-d–400	14	472.4	402.4	678.3	402.4	SS	82167.2	6.54
B-250-d–500	14	562.1	402.4	880.8	402.4	SS	100101.7	7.80
B-400-d–200	6	201.3	1938.7	277.2	201.3	CB	39667.39	0.43
B-400-d–400	12	755.8	1938.7	847.7	755.8	CB	168516.4	1.80
B-400-d–600	26	1600.6	1938.7	2070.5	1600.6	CB	315822.0	3.37
B-400-d–800	26	2255.4	1938.7	2668.8	1938.7	SS	423150.4	4.52
B-250-n–0.2	10	386.1	399.1	560.8	386.1	CB	95364.0	7.60
B-250-n–0.4	10	489.8	299.3	701.0	299.3	SS	101627.5	8.10
B-400-n–0.2	8	867.0	1826.7	1074.2	867.0	CB	250829.9	2.68
B-400-n–0.4	8	1246.2	1370.1	1543.8	1246.2	CB	296796.3	3.20
B-550-n–0.2	8	1593.9	4748.8	1928.7	1593.9	CB	453396.9	1.40
B-550-n–0.4	8	2448.6	3561.6	2900.0	2448.6	CB	573946.3	1.71
B-250-f_{cu}–35	10	274.1	402.4	401.0	274.1	CB	57720.6	4.60
B-250-f_{cu}–50	10	391.6	413.0	442.4	391.6	CB	66581.9	5.20
B-250-f_{cu}–60	10	420.7	421.6	472.9	420.7	CB	72007.6	5.53
B-400-f_{cu}–35	8	438.6	1938.7	489.8	438.6	CB	96826.3	1.03
B-400-f_{cu}–50	8	626.6	1978.3	553.0	553.0	UPS	117476.2	1.24
B-400-f_{cu}–60	8	673.1	2011.2	601.9	601.9	UPS	137918.1	1.43
B-550-f_{cu}–35	8	603.1	5039.9	647.3	603.1	CB	124142.2	0.37
B-550-f_{cu}–50	8	861.6	5142.8	738.5	738.5	UPS	155672.7	0.46
B-550-f_{cu}–60	8	925.5	5228.4	809.7	809.7	UPS	178893.6	0.52
B-250-α_c–0.6	10	274.1	403.8	401.0	274.1	CB	59331.7	4.56
B-250-α_c–0.7	10	274.1	413.0	401.0	274.1	CB	58885.3	4.60
B-400-α_c–0.6	8	438.6	1943.9	489.8	438.6	CB	97835.5	1.02
B-400-α_c–0.7	8	438.6	1978.3	489.8	438.6	CB	97285.0	1.02
B-550-α_c–0.6	8	603.1	5053.4	647.3	603.1	CB	124086.6	0.36
B-550-α_c–0.7	8	603.1	5142.8	647.3	603.1	CB	123061.4	0.36
B-250-W_b-1020	8	274.1	402.4	350.7	274.1	CB	58998.3	4.70
B-250-W_b-1220	8	274.1	402.4	350.7	274.1	CB	59142.6	4.71
B-250-W_b-1420	8	274.1	402.4	350.7	274.1	CB	59016.5	4.70
B-400-W_b-1220	8	438.6	1938.7	489.8	438.6	CB	96826.3	1.03
B-400-W_b-1620	8	438.6	1938.7	489.8	438.6	CB	97674.5	1.04
B-400-W_b-2220	8	438.6	1938.7	489.8	438.6	CB	98117.7	1.05

Note: CB - Concrete bearing failure; SS - Steel section failure; UPS - Upward punching shear failure.

GB50017 [13], the required embedment depth is provided by assuming M_c not less than M_y to insure the connection not being the weak point. Based on the similar assumption, the required embedment depth of the embedded CFST column base connection (d_{req}) using the proposed solution is also developed:

$$d_{req} \geq \frac{V}{f_{bearing}B} + \sqrt{2\left(\frac{V}{f_{bearing}B}\right)^2 + \frac{4M}{f_{bearing}B}} \tag{35}$$

Table 5
Comparison of numerical and theoretical bending resistances.

Specimens	M_{FEM} (kN·m)	Existing methods				$\frac{M_{The}}{M_{FEM}}$	$\frac{M_{AISC}}{M_{FEM}}$	$\frac{M_{GB}}{M_{FEM}}$	$\frac{M_{JGJ}}{M_{FEM}}$	$\frac{M_{Grilli}}{M_{FEM}}$
		M_{AISC} (kN·m)	M_{GB} (kN·m)	M_{JGJ} (kN·m)	M_{Grilli} (kN·m)					
B-250-*d*−100	38.4	36.8	14.2	9.4	23.9	0.85	0.96	0.37	0.24	0.62
B-250-*d*−200	134.7	139.4	54.8	37.5	92.5	0.93	1.04	0.41	0.28	0.69
B-250-*d*−300	299.3	298.1	119.5	84.3	201.6	0.92	1.00	0.40	0.28	0.67
B-250-*d*−400	482.9	504.8	205.8	149.8	347.4	0.83*	1.05	0.43	0.31	0.72
B-250-*d*−500	586.7	752.9	311.9	234.1	477.7	0.69*	1.28	0.53	0.40	0.81
B-400-*d*−200	196.4	213.8	87.7	59.9	147.9	1.02	1.09	0.45	0.31	0.75
B-400-*d*−400	762.3	774.0	329.3	239.7	555.8	0.99	1.02	0.43	0.31	0.73
B-400-*d*−600	1619.4	1590.3	697.4	539.3	1177.7	0.99	0.98	0.43	0.33	0.73
B-400-*d*−800	2525.0	2601.4	1169.9	958.8	1976.8	0.77*	1.03	0.46	0.38	0.78
B-250-*n*−0.2	437.3	298.1	119.5	84.3	201.6	0.88	0.68	0.27	0.19	0.46
B-250-*n*−0.4	450.8	298.1	119.5	84.3	201.6	0.66*	0.66	0.26	0.19	0.45
B-400-*n*−0.2	1033.3	457.1	191.1	134.8	322.5	0.84	0.44	0.18	0.13	0.31
B-400-*n*−0.4	1263.6	457.1	191.1	134.8	322.5	0.99	0.36	0.15	0.11	0.26
B-550-*n*−0.2	1855.8	615.8	262.8	185.4	443.4	0.86	0.33	0.14	0.10	0.24
B-550-*n*−0.4	2450.4	615.8	262.8	185.4	443.4	1.00	0.25	0.11	0.08	0.18
B-250-f_{cu}−35	299.3	298.1	119.5	84.3	201.6	0.92	1.00	0.40	0.28	0.67
B-250-f_{cu}−50	375.9	356.3	170.6	120.4	287.9	1.04	0.95	0.45	0.32	0.77
B-250-f_{cu}−60	423.3	400.8	196.5	138.6	364.3	0.99	0.95	0.46	0.33	0.86
B-400-f_{cu}−35	443.7	457.1	191.1	134.8	322.5	0.99	1.03	0.43	0.30	0.73
B-400-f_{cu}−50	516.5	546.3	273.0	192.6	460.7	1.07	1.06	0.53	0.37	0.89
B-400-f_{cu}−60	608.0	614.5	314.4	221.8	582.9	0.99	1.01	0.52	0.36	0.96
B-550-f_{cu}−35	617.5	615.8	262.8	185.4	443.4	0.98	1.00	0.43	0.30	0.72
B−550-f_{cu}−50	720.3	736.1	375.4	264.8	633.5	1.03	1.02	0.52	0.37	0.88
B-550-f_{cu}−60	821.2	828.0	432.2	304.9	801.5	0.99	1.01	0.53	0.37	0.98
B-250-α_c−0.6	302.6	298.1	119.5	84.3	201.6	0.91	0.99	0.39	0.28	0.67
B-250-α_c−0.7	301.6	298.1	119.5	84.3	201.6	0.91	0.99	0.40	0.28	0.67
B-400-α_c−0.6	454.6	457.1	191.1	134.8	322.5	0.96	1.01	0.42	0.30	0.71
B-400-α_c−0.7	446.1	457.1	191.1	134.8	322.5	0.98	1.02	0.43	0.30	0.72
B-550-α_c−0.6	623.5	615.8	262.8	185.4	443.4	0.97	0.99	0.42	0.30	0.71
B-550-α_c−0.7	618.2	615.8	262.8	185.4	443.4	0.98	1.00	0.43	0.30	0.72
B-250-W_b-1020	316.7	344.3	119.5	84.3	201.6	0.87	1.09	0.37	0.26	0.62
B-250-W_b-1220	324.4	387.5	119.5	84.3	201.6	0.85	1.19	0.36	0.25	0.61
B-250-W_b-1420	331.3	428.3	119.5	84.3	201.6	0.83	1.29	0.36	0.25	0.61
B-400-W_b-1220	443.8	454.6	191.1	134.8	322.5	0.99	1.02	0.42	0.30	0.71
B-400-W_b-1620	451.3	548.2	191.1	134.8	322.5	0.97	1.21	0.42	0.30	0.71
B-400-W_b-2220	455.7	674.9	191.1	134.8	322.5	0.96	1.48	0.40	0.28	0.68
Avg.						0.95^{+}	0.96	0.39	0.28	0.65
Cov.						0.07^{+}	0.27	0.27	0.28	0.33

Note: * Failed by CFST section; + Steel section failure cases are not included.

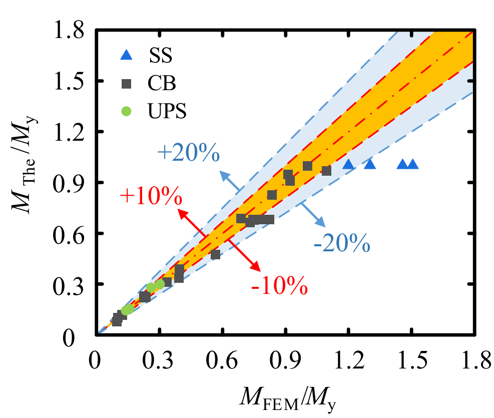


Fig. 27. Performance of the proposed solution.

where the bending moment M and shear force V are about the reference section. It is worth mentioning that the axial force N is not considered in Eq. (35), which is due to the axial force is beneficial to the ultimate load of the embedded CFST column base connection, according to the above discussion.

By assuming the height of the column's inflection point being 1.5 m from the reference section (i.e., the story height is 3.0 m), the d_{req} values for all examples are given in Table 6. As illustrated in Table 6, d_{req} deduced based of the proposed solution are much smaller than those calculated using the method of GB50017 [13], JGJ 138 [15] and JGJ 99 [16], being approximately 0.63, 0.58, and 0.61 times of these three methods on average. This indicates that the embedment depth of the embedded CFST column base connection can be significantly reduced. As illustrated in Table 6, the generated d_{req} values may also slightly smaller than d_{eff} for all examples.

6. Concluding remarks

The seismic performance of embedded CFST column base connections are investigated in this study. Based on the study of this paper, the following conclusions may be drawn:

(1) For the embedded CFST column base connections with small embedment depths, the crushing of the surrounding concrete along the loading direction is significant, which may lead to the sliding phenomenon in the hysteresis curve.
(2) According the test and FE analyses, the embedded CFST column base connections may fail by the concrete bearing failure, the upward punching failure and the CFST column failure.
(3) The axial force may be beneficial to the ultimate load of the embedded CFST column base connection when its failure is governed by the concrete bearing failure, the upward punching failure.
(4) The proposed solution, with N, V and M accounted, performs well in predicting the ultimate load of the embedded CFST column

Table 6
Comparison of required embedment depth.

Specimen	M_y (kN·m)	d_{eff} (mm)	d_{req} (mm)	GB 50017 (mm)	JGJ 138 (mm)	JGJ 99 (mm)
B-250-d−100	402.4	439.0	367.3	574.3	655.6	625.0
B-250-d−200	402.4	439.0	367.3	574.3	655.6	625.0
B-250-d−300	402.4	439.0	367.3	574.3	655.6	625.0
B-250-d−400	402.4	439.0	367.3	574.3	655.6	625.0
B-250-d−500	402.4	439.0	367.3	574.3	655.6	625.0
B-400-d−200	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-d−400	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-d−600	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-d−800	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-250-f_{cu}−35	402.4	439.0	367.3	574.3	655.6	625.0
B-250-f_{cu}−50	413.0	422.0	308.5	479.9	555.6	625.0
B-250-f_{cu}−60	421.6	411.3	300.3	449.8	523.2	625.0
B-400-f_{cu}−35	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-f_{cu}−50	1978.3	696.4	554.3	880.3	961.4	1000.0
B-400-f_{cu}−60	2011.2	678.1	537.9	820.3	903.5	1000.0
B-550-f_{cu}−35	5039.9	997.5	955.2	1565.6	1564.2	1375.0
B−550-f_{cu}−50	5142.8	957.5	788.7	1276.4	1322.0	1375.0
B-550-f_{cu}−60	5228.4	932.4	764.7	1185.3	1242.3	1375.0
B-250-α_c−0.6	403.8	439.3	368.0	575.4	656.7	625.0
B-250-α_c−0.7	413.0	441.2	372.4	582.5	664.1	625.0
B-400-α_c−0.6	1943.9	725.8	667.9	1070.5	1139.1	1000.0
B-400-α_c−0.7	1978.3	728.1	674.4	1081.5	1149.1	1000.0
B-550-α_c−0.6	5053.4	998.0	956.7	1568.2	1566.3	1375.0
B-550-α_c−0.7	5142.8	1001.2	966.4	1585.3	1580.1	1375.0
B-250-W_b-1020	402.4	439.0	367.3	574.3	655.6	625.0
B-250-W_b-1220	402.4	439.0	367.3	574.3	655.6	625.0
B-250-W_b-1420	402.4	439.0	367.3	574.3	655.6	625.0
B-400-W_b-1220	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-W_b-1620	1938.7	725.4	666.9	1068.8	1137.6	1000.0
B-400-W_b-2220	1938.7	725.4	666.9	1068.8	1137.6	1000.0

base connections, and it is superior to the existing solutions in design codes.

(5) The required embedment depth for the embedded CFST column base connection based on the proposed method is smaller than the requirements of GB 50017, JGJ 138 and JGJ 99, which may bring many conveniences to in-site installation in practical use.

(6) Table 7 summarizes the effects of main parameters on the behaviors of embedded CFST column base connections for the reference of design.

Table 7
Summary of key parameters and proposed formulae.

Parameters	Explanation	Simple description
d_{embed}	Embedment depth	Important parameter for engineering design. Significant effects on CB resistance.
n	Axial load ratio	Compression is beneficial to CB resistance, but may reduce the section strength of CFST column.
f_{cu}	Strength of concrete foundation	Increasing f_{cu} may improve both resistance and stiffness of column base governed by CB or UPS failures.
α_c	Concrete strength ratio	Negligible effect on CB resistance.
W_b	Width of concrete foundation	W_b/B not less than 3.0 is recommended
CB	Concrete bearing failure	Main controlling mode for shallowly embedded column base. CB resistance is given in Eq. 30.
UPS	Upward punching failure	Uncommon controlling mode for shallowly embedded column base. Only occurs when the relatively high strength concrete of foundation is used. UPS resistance is given in Eq. 33.
d_{req}	Required embedment depth	Given in Eq. 35 (based on CB resistance=CFST section)

CRediT authorship contribution statement

Hu Zhongzheng: Writing – review & editing, Supervision, Conceptualization. **Tong Genshu:** Writing – review & editing, Supervision. **Zhang Lei:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Liu Yufei:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

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